TSIA2 Math Practice Test 15

PART A — QUESTIONS

TSIA2-M15-001
Section: Math
Type: mcq
Category: Quantitative Reasoning
Prompt:
A student compares two ways to get a textbook for one semester.

Option A: Buy the textbook for \$96 and sell it back for \$42 at the end of the semester.
Option B: Rent the textbook for \$17 per month for 4 months and pay a \$5 processing fee.

How much less does the cheaper option cost?
A) \$9
B) \$15
C) \$19
D) \$23

TSIA2-M15-002
Section: Math
Type: mcq
Category: Algebraic Reasoning
Prompt:
The formula $D = rt + s$ gives the total distance D traveled after starting s miles from a checkpoint and traveling at rate r for t hours. Which equation correctly solves the formula for r?
A) $r = D - s/t$
B) $r = (D - s)/t$
C) $r = D/(t + s)$
D) $r = t(D - s)$

TSIA2-M15-003
Section: Math
Type: mcq
Category: Geometric and Spatial Reasoning
Prompt:
A community garden has the shape of a large rectangle that is 18 feet by 12 feet. A square seating area with side length 5 feet is removed from one corner. What is the remaining planting area?
A) 143 square feet
B) 181 square feet
C) 191 square feet
D) 241 square feet

TSIA2-M15-004
Section: Math
Type: mcq
Category: Probabilistic and Statistical Reasoning
Prompt:
The table shows the results of a survey of 120 students.

Uses campus gym and lives on campus: 32
Uses campus gym and lives off campus: 28
Does not use campus gym and lives on campus: 24
Does not use campus gym and lives off campus: 36

If one student who lives on campus is selected at random, what is the probability that the student uses the campus gym?
A) 4/15
B) 8/15
C) 4/7
D) 1/2

TSIA2-M15-005
Section: Math
Type: mcq
Category: Quantitative Reasoning
Prompt:
A medical assistant prepares a solution using 1.8 milliliters of concentrate for every 12 milliliters of water. At the same rate, how many milliliters of concentrate are needed for 50 milliliters of water?
A) 6.5 milliliters
B) 7.0 milliliters
C) 7.5 milliliters
D) 8.0 milliliters

TSIA2-M15-006
Section: Math
Type: mcq
Category: Algebraic Reasoning
Prompt:
A linear function is shown in the table.

x: 3, 6, 9, 12
y: 11, 17, 23, 29

Which equation represents the relationship between x and y?
A) $y = 2x + 5$
B) $y = 3x + 2$
C) $y = 6x - 7$
D) $y = x + 8$

TSIA2-M15-007
Section: Math
Type: mcq
Category: Geometric and Spatial Reasoning
Prompt:
A right rectangular pyramid has a square base with side length 10 inches. A cross-section parallel to the base is similar to the base and has side length 4 inches. What is the area of the cross-section?
A) 8 square inches
B) 16 square inches
C) 40 square inches
D) 100 square inches

TSIA2-M15-008
Section: Math
Type: mcq
Category: Probabilistic and Statistical Reasoning
Prompt:
Two stores recorded the number of customers entering during five one-hour periods.

Store A: 42, 44, 45, 46, 48
Store B: 30, 38, 45, 52, 60

Which statement best compares the two data sets?
A) Store A and Store B have the same median, but Store A has a smaller range.
B) Store A and Store B have the same median, but Store B has a smaller range.
C) Store A has a greater median and a greater range than Store B.
D) Store B has a greater median and a smaller range than Store A.

TSIA2-M15-009
Section: Math
Type: mcq
Category: Quantitative Reasoning
Prompt:
A school orders 48 tablets. Each tablet costs $215. The school receives a discount of $30 per tablet. What is the total cost after the discount?
A) $8,880
B) $9,360
C) $10,320
D) $11,760

TSIA2-M15-010
Section: Math
Type: mcq
Category: Algebraic Reasoning
Prompt:
Two landscaping companies charge for the same type of yard work.

Company A: $120 equipment fee plus $35 per hour
Company B: $60 equipment fee plus $50 per hour

For how many hours will the total charge be the same for both companies?
A) 3 hours
B) 4 hours
C) 5 hours
D) 6 hours

TSIA2-M15-011
Section: Math
Type: mcq
Category: Geometric and Spatial Reasoning
Prompt:
A line segment has endpoints $(-5, -1)$ and $(7, 4)$. Which value is closest to the length of the segment?

A) 11 units
B) 12 units
C) 13 units
D) 17 units

TSIA2-M15-012
Section: Math
Type: mcq
Category: Probabilistic and Statistical Reasoning
Prompt:
A final grade is calculated using 25% homework, 35% quizzes, and 40% exams. A student's averages are 92 for homework, 84 for quizzes, and 78 for exams. What is the student's final grade?
A) 81.6
B) 83.6
C) 84.8
D) 86.4

TSIA2-M15-013
Section: Math
Type: mcq
Category: Quantitative Reasoning
Prompt:
A community college reports that 420 students attended a career fair. This was 28% of the students invited. How many students were invited?
A) 1,176
B) 1,260
C) 1,500
D) 1,680

TSIA2-M15-014
Section: Math
Type: mcq
Category: Algebraic Reasoning
Prompt:
A population of cells is modeled by $P(t) = 600(1.15)^t$, where t is the number of hours. What does the factor 1.15 mean in this model?
A) The population increases by 15% each hour.
B) The population increases by 115 cells each hour.
C) The population starts with 1.15 cells.
D) The population decreases by 15% each hour.

TSIA2-M15-015
Section: Math
Type: mcq
Category: Geometric and Spatial Reasoning
Prompt:
Two similar triangles have corresponding side lengths in a ratio of 4:9. The area of the smaller triangle is 48 square centimeters. What is the area of the larger triangle?
A) 108 square centimeters
B) 192 square centimeters

C) 243 square centimeters
D) 972 square centimeters

TSIA2-M15-016
Section: Math
Type: mcq
Category: Probabilistic and Statistical Reasoning
Prompt:
A workshop organizer tracks registrations over four days.

Day 1: 36 registrations
Day 2: 44 registrations
Day 3: 53 registrations
Day 4: 63 registrations

Which conclusion is best supported by the data?
A) Registrations increased by the same number each day.
B) Registrations increased each day, but not by a constant amount.
C) Registrations decreased after Day 2.
D) The total number of registrations was less than 180.

TSIA2-M15-017
Section: Math
Type: mcq
Category: Quantitative Reasoning
Prompt:
A rectangular field measures 49.6 meters by 20.2 meters. Which is the best estimate of the field's area?
A) 800 square meters
B) 900 square meters
C) 1,000 square meters
D) 1,200 square meters

TSIA2-M15-018
Section: Math
Type: mcq
Category: Algebraic Reasoning
Prompt:
The height, in feet, of a thrown object is modeled by $h(t) = -16t^2 + 32t + 6$. What is the height of the object after 2 seconds?
A) 6 feet
B) 10 feet
C) 22 feet
D) 38 feet

TSIA2-M15-019
Section: Math
Type: mcq
Category: Geometric and Spatial Reasoning
Prompt:
Two parallel lines are cut by a transversal. One angle measures 74°. An angle on the same side of

the transversal and inside the parallel lines is paired with it. What is the measure of that angle?
A) 16°
B) 74°
C) 106°
D) 148°

TSIA2-M15-020
Section: Math
Type: mcq
Category: Probabilistic and Statistical Reasoning
Prompt:
A data set contains the values 22, 24, 25, 27, 28, and 30. If the value 72 is added to the data set, which statement is true?
A) The median decreases, and the mean increases.
B) The mean increases, and the range increases.
C) The mean stays the same, and the range increases.
D) The median increases more than the range increases.

PART B — ANSWER KEY + EXPLANATIONS

TSIA2-M15-001 — Correct: C — Correct Answer: \$19 — Explanation: Option A costs \$96 − \$42 = \$54. Option B costs 4(\$17) + \$5 = \$73. The cheaper option is \$73 − \$54 = \$19 less.
TSIA2-M15-002 — Correct: B — Correct Answer: $r = (D - s)/t$ — Explanation: Starting with $D = rt + s$, subtract s from both sides to get $D - s = rt$. Divide by t to get $r = (D - s)/t$.
TSIA2-M15-003 — Correct: C — Correct Answer: 191 square feet — Explanation: The large rectangle has area 18 × 12 = 216 square feet. The removed square has area 5 × 5 = 25 square feet. The remaining area is 216 − 25 = 191 square feet.
TSIA2-M15-004 — Correct: C — Correct Answer: 4/7 — Explanation: There are 32 + 24 = 56 students who live on campus. Of those, 32 use the gym. The probability is 32/56 = 4/7.
TSIA2-M15-005 — Correct: C — Correct Answer: 7.5 milliliters — Explanation: The concentrate-to-water ratio is 1.8/12 = 0.15. For 50 milliliters of water, the concentrate needed is 50 × 0.15 = 7.5 milliliters.
TSIA2-M15-006 — Correct: A — Correct Answer: $y = 2x + 5$ — Explanation: When x increases by 3, y increases by 6, so the slope is 2. Using $x = 3$ and $y = 11$ gives $11 = 2(3) + 5$.
TSIA2-M15-007 — Correct: B — Correct Answer: 16 square inches — Explanation: The cross-section is a square with side length 4 inches, so its area is $4^2 = 16$ square inches.
TSIA2-M15-008 — Correct: A — Correct Answer: Store A and Store B have the same median, but Store A has a smaller range. — Explanation: Both medians are 45. Store A's range is 48 − 42 = 6, and Store B's range is 60 − 30 = 30.
TSIA2-M15-009 — Correct: A — Correct Answer: \$8,880 — Explanation: The discounted price per tablet is \$215 − \$30 = \$185. For 48 tablets, the total cost is 48 × \$185 = \$8,880.
TSIA2-M15-010 — Correct: B — Correct Answer: 4 hours — Explanation: Set the costs equal: $120 + 35h = 60 + 50h$. Then $60 = 15h$, so $h = 4$.
TSIA2-M15-011 — Correct: C — Correct Answer: 13 units — Explanation: The horizontal change is $7 - (-5) = 12$, and the vertical change is $4 - (-1) = 5$. The distance is $\sqrt{(12^2 + 5^2)} = \sqrt{169} = 13$ units.
TSIA2-M15-012 — Correct: B — Correct Answer: 83.6 — Explanation: The weighted grade is 0.25(92) + 0.35(84) + 0.40(78) = 23 + 29.4 + 31.2 = 83.6.
TSIA2-M15-013 — Correct: C — Correct Answer: 1,500 — Explanation: If 420 is 28% of the invited students, then the number invited is 420 ÷ 0.28 = 1,500.
TSIA2-M15-014 — Correct: A — Correct Answer: The population increases by 15% each hour. —

Explanation: Multiplying by 1.15 means the population keeps 100% of its previous size and gains 15% each hour.
TSIA2-M15-015 — Correct: C — Correct Answer: 243 square centimeters — Explanation: The length scale factor from the smaller triangle to the larger triangle is 9/4. Area scales by $(9/4)^2 = 81/16$. The larger area is $48 \times 81/16 = 243$ square centimeters.
TSIA2-M15-016 — Correct: B — Correct Answer: Registrations increased each day, but not by a constant amount. — Explanation: Registrations increased by 8, then 9, then 10. The values increase each day, but the amount of increase is not constant.
TSIA2-M15-017 — Correct: C — Correct Answer: 1,000 square meters — Explanation: Estimate 49.6 as 50 and 20.2 as 20. Then $50 \times 20 = 1{,}000$ square meters.
TSIA2-M15-018 — Correct: A — Correct Answer: 6 feet — Explanation: Substitute $t = 2$: $h(2) = -16(2^2) + 32(2) + 6 = -64 + 64 + 6 = 6$ feet.
TSIA2-M15-019 — Correct: C — Correct Answer: 106° — Explanation: Same-side interior angles formed by parallel lines are supplementary, so the paired angle is $180° - 74° = 106°$.
TSIA2-M15-020 — Correct: B — Correct Answer: The mean increases, and the range increases. — Explanation: Adding 72 increases the total, so the mean increases. It also changes the maximum from 30 to 72, so the range increases.